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1. Experiment Title:

Implement localStorage and sessionStorage to save user preferences like theme settings.

2. Software/Tools Required:

Google Chrome (Web Browser), Visual Studio Code (Code Editor), Browser Developer Tools / Console, HTML, CSS, Javascript

3. Experiment Program Code:

Experiment 9.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Theme Preferences</title>
    <link rel="stylesheet" href="Experiment 9.css">
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700&display=swap"
rel="stylesheet">
</head>
<body>
    <div class="container">
        <h2>Theme Preferences</h2>
        <p class="subtitle">Using localStorage and sessionStorage to save user preferences.</p>

        <div class="button-group">
            <button id="lightBtn" onclick="setTheme('light')">Light Theme</button>
            <button id="darkBtn" onclick="setTheme('dark')">Dark Theme</button>
            <button id="clearBtn" onclick="clearPreference()">Clear Preference</button>
        </div>

        <footer>
            Made by Rohan Laharwani PRN - 24070521055
        </footer>
    </div>

    <script src="Experiment 9.js"></script>
</body>
</html>
```

Experiment 9.css

```
* {
    box-sizing: border-box;
}

body {
    background-color: whitesmoke;
    color: #333;
    font-family: 'Inter', sans-serif;
    margin: 0;
    padding: 40px 20px;
```

```
    display: flex;
    justify-content: center;
    align-items: center;
    min-height: 100vh;
    transition: background-color 0.3s ease, color 0.3s ease;
}

body.dark {
    background-color: #1e1e2e;
    color: #cdd6f4;
}

.container {
    width: 100%;
    max-width: 500px;
    background-color: white;
    padding: 35px 30px;
    border: 1px solid lightgray;
    border-radius: 12px;
    box-shadow: 0 4px 15px lightgray;
    text-align: center;
    transition: background-color 0.3s ease, border-color 0.3s ease, box-shadow 0.3s ease;
}

body.dark .container {
    background-color: #252739;
    border-color: #3e4259;
    box-shadow: 0 4px 15px rgba(0, 0, 0, 0.4);
}

h2 {
    color: darkblue;
    margin-top: 0;
    margin-bottom: 12px;
    font-size: 24px;
    transition: color 0.3s ease;
}

body.dark h2 {
    color: #89b4fa;
}

.subtitle {
    font-size: 14px;
    color: #555;
    margin-bottom: 24px;
    line-height: 1.5;
    transition: color 0.3s ease;
}

body.dark .subtitle {
    color: #a6adc8;
}

.button-group {
    display: flex;
    justify-content: center;
    gap: 10px;
    flex-wrap: wrap;
    margin-bottom: 10px;
}

button {
    padding: 10px 18px;
    background-color: royalblue;
    color: white;
    border: none;
    border-radius: 6px;
    font-size: 14px;
```

```
    font-weight: 600;
    cursor: pointer;
    transition: background-color 0.2s, transform 0.1s ease;
}

button:hover {
    background-color: blue;
}

button:active {
    transform: scale(0.98);
}

#clearBtn {
    background-color: #6c757d;
}

#clearBtn:hover {
    background-color: #5a6268;
}

footer {
    text-align: center;
    margin-top: 30px;
    font-size: 14px;
    color: gray;
    border-top: 1px solid lightgray;
    padding-top: 15px;
    transition: border-color 0.3s ease, color 0.3s ease;
}

body.dark footer {
    color: #a6adc8;
    border-top-color: #3e4259;
}
```

Experiment 9.js

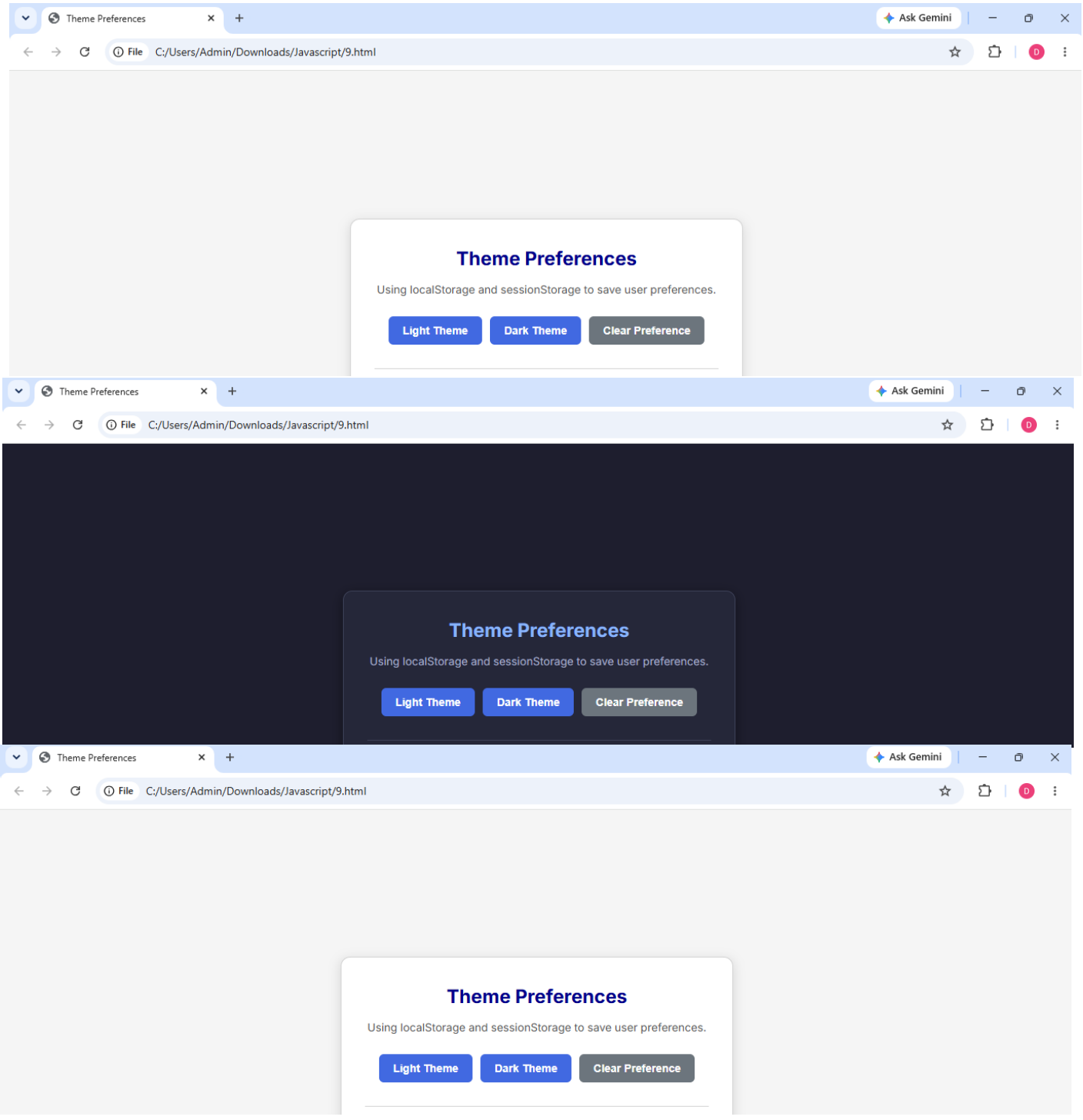
```
function setTheme(theme) {
    if (theme === 'dark') {
        document.body.classList.add('dark');
    } else {
        document.body.classList.remove('dark');
    }

    localStorage.setItem('theme', theme);
    sessionStorage.setItem('sessionTheme', theme);
}

function clearPreference() {
    localStorage.removeItem('theme');
    sessionStorage.removeItem('sessionTheme');
    document.body.classList.remove('dark');
}

window.onload = function () {
    let saved = localStorage.getItem('theme');
    if (saved === 'dark') {
        document.body.classList.add('dark');
    }
};
```

4. Output:


Theme Preferences
Ask Gemini
File C:/Users/Admin/Downloads/Javascript/9.html
Theme Preferences
Using localStorage and sessionStorage to save user preferences.
Light Theme
Dark Theme
Clear Preference

5. Result/Conclusion:

The Web Storage API was implemented to persist user preferences such as theme settings using localStorage.setItem(), getItem() and removeItem(), allowing a Dark/Light mode toggle to remain consistent across browser reloads. This reinforced the use of localStorage and sessionStorage for client-side data persistence in JavaScript.